



Software Development

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BorrowBondhu – A Peer-to-Peer Rental Marketplace Platform

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BorrowBondhu – A Peer-to-Peer Rental Marketplace Platform

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the complete functional and non-functional requirements for **BorrowBondhu**, a web-based Peer-to-Peer (P2P) rental marketplace platform.

BorrowBondhu enables users in Bangladesh to rent unused personal items such as projectors, cameras, construction tools, and event equipment to others who require them for short-term use.

This document serves as a reference for developers, project supervisors, and stakeholders to understand system functionality, constraints, and operational behavior.

1.2 Scope

BorrowBondhu will be developed initially as a Minimum Viable Product (MVP). The system will support core marketplace functionalities including user registration and authentication, listing management, search and browsing, booking management, review and rating mechanisms, and administrative supervision.

Features such as online payment gateway integration, mobile application development, delivery logistics integration, security deposit handling, and NID-based identity verification are considered future enhancements and are outside the scope of the MVP phase.

The system aims to promote digital sharing economy practices and reduce unnecessary purchases by enabling resource sharing.

1.3 Definitions and Acronyms

Term	Definition
MVP	Minimum Viable Product
P2P	Peer-to-Peer
User	Registered individual (Lender or Renter)
Admin	Platform administrator
Listing	Item available for rent
Booking	Rental request and confirmation process
NID	National Identity Card

1.4 References

- Project Proposal – BorrowBondhu (Group 10)
- System Architecture Plan:
 - Frontend: HTML/CSS/JavaScript or React
 - Backend: Node.js with Express / Django / GoLang
 - Database: MongoDB / PostgreSQL
 - Hosting: Firebase / Vercel / Local Server

2. Overall Description

2.1 Product Perspective

BorrowBondhu is an independent web-based system designed to address the absence of a trusted equipment rental platform in Bangladesh. The system operates as a centralized digital marketplace connecting renters and lenders.

The platform does not directly manage physical delivery or payments in its MVP phase. Instead, it facilitates communication, booking confirmation, and trust building through reviews and ratings.

2.2 Product Functions

The system will allow users to:

- Register and log in securely
- Manage personal profiles
- Create, edit, and delete item listings
- Search and browse available items
- Request and manage bookings
- Leave reviews and ratings
- Allow admin monitoring and dispute handling

2.3 User Characteristics

2.3.1 University Students

- Age: 18–25

- Tech savvy
- Need short-term equipment access
- Prefer affordable and simple systems

2.3.2 Event Organizers

- Age: 25 – 40
- Require reliability and quick booking
- Moderate technical skills

2.3.3 Freelancers (Photographers, Designers, etc.)

- Digitally literate
- May act as both lender and renter
- Value reviews and trust

2.3.4 Small Businesses

- Basic technical knowledge
- Need efficient filtering and availability tracking

2.3.5 General Public

- Mixed technical skills
- Require intuitive user interface
- Access via smartphones or desktops

2.4 Operating Environment

BorrowBondhu will operate as a web based application accessible through modern browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The frontend will be developed using HTML, CSS, JavaScript, or React. The backend will utilize Node.js with Express or Django/GoLang. Data will be stored in MongoDB or PostgreSQL databases. The application will be deployed using platforms such as Firebase, Vercel, or similar hosting environments.

2.5 Design and Implementation Constraints

- Use open-source technologies
- Compliance with Bangladeshi digital laws
- No integrated payment system in MVP
- Web-based application only

2.6 Assumptions and Dependencies

- Users have valid email and phone number
- Internet access is available
- Hosting services (Firebase/Vercel) are operational
- Users act responsibly and legally

3. System Modules

3.1 User Authentication Module

- Registration
- Login/Logout
- Password reset
- Secure session management

3.2 Listing Management Module

- Upload listings
- Edit/Delete listings
- Set price and availability

3.3 Search and Browsing Module

- Keyword search
- Category filter
- Price filter
- View item details

3.4 Booking and Availability Module

- Send booking request
- Accept/Reject booking
- Calendar based availability

3.5 Review and Rating Module

- Post booking review
- 1 – 5 star rating
- Display average rating

3.6 Admin Dashboard Module

- User management
- Listing approval
- Dispute handling
- Commission tracking

3.7 Database Module

- Store users
- Store listings
- Store bookings
- Store reviews

3.8 Frontend UI Module

- Responsive web interface
- Forms and dashboards

4. Functional Requirements

Functional requirements are specified using structured requirement statements.

FR-1: The system shall allow new users to register by providing name, email, password, and phone number.

FR-2: The system shall validate the uniqueness of email and phone number during registration.

FR-3: The system shall authenticate registered users using secure login mechanisms.

FR-4: The system shall allow users to update their profile information after successful login.

FR-5: The system shall allow authenticated users to create item listings including title, description, rental price, images, and availability schedule.

FR-6: The system shall store submitted listings and require administrative approval before publication.

FR-7: The system shall allow listing owners to edit or delete their listings.

FR-8: The system shall allow users to search listings using keywords and filter criteria.

FR-9: The system shall allow renters to submit booking requests specifying rental dates.

FR-10: The system shall notify lenders when a booking request is submitted.

FR-11: The system shall allow lenders to accept or reject booking requests.

FR-12: The system shall update booking status based on lender decisions.

FR-13: The system shall allow users to submit ratings and written reviews after a completed booking.

FR-14: The system shall restrict review submission to users involved in verified bookings.

FR-15: The system shall allow administrators to suspend or activate user accounts.

FR-16: The system shall allow administrators to approve or reject listings.

FR-17: The system shall record commission data for completed bookings.

5. Non-Functional Requirements

5.1 Performance

- Page load time ≤ 2 seconds
- Support 1000 concurrent users
- 99% uptime

5.2 Security

- HTTPS encryption
- Password hashing
- Secure session tokens
- Future 2FA option

5.3 Usability

- Mobile responsive design
- Simple navigation
- Bangla language support

5.4 Reliability

- Graceful error handling
- Daily database backup

5.5 Maintainability

- MVC architecture
- Modular code
- API documentation

5.6 Portability

- Cross-browser compatibility
- Cloud deployment ready

6. External Interface Requirements

The user interface shall consist of web based graphical forms, dashboards, and search components.

The software interface shall include APIs for database operations and potential integration with external services in future phases.

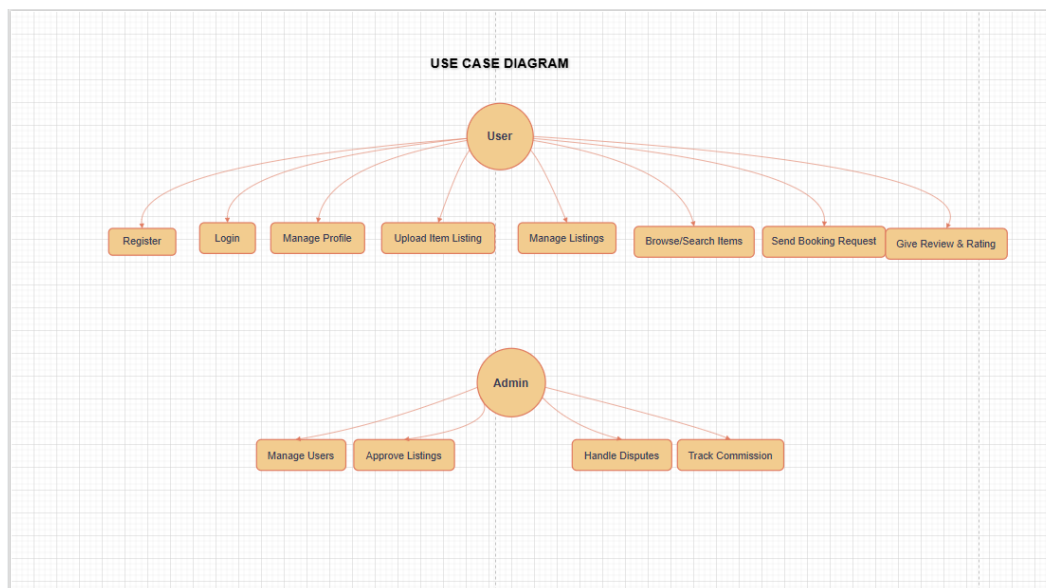
Communication interfaces shall support email or SMS notifications for booking confirmations and account updates.

7. Other Requirements

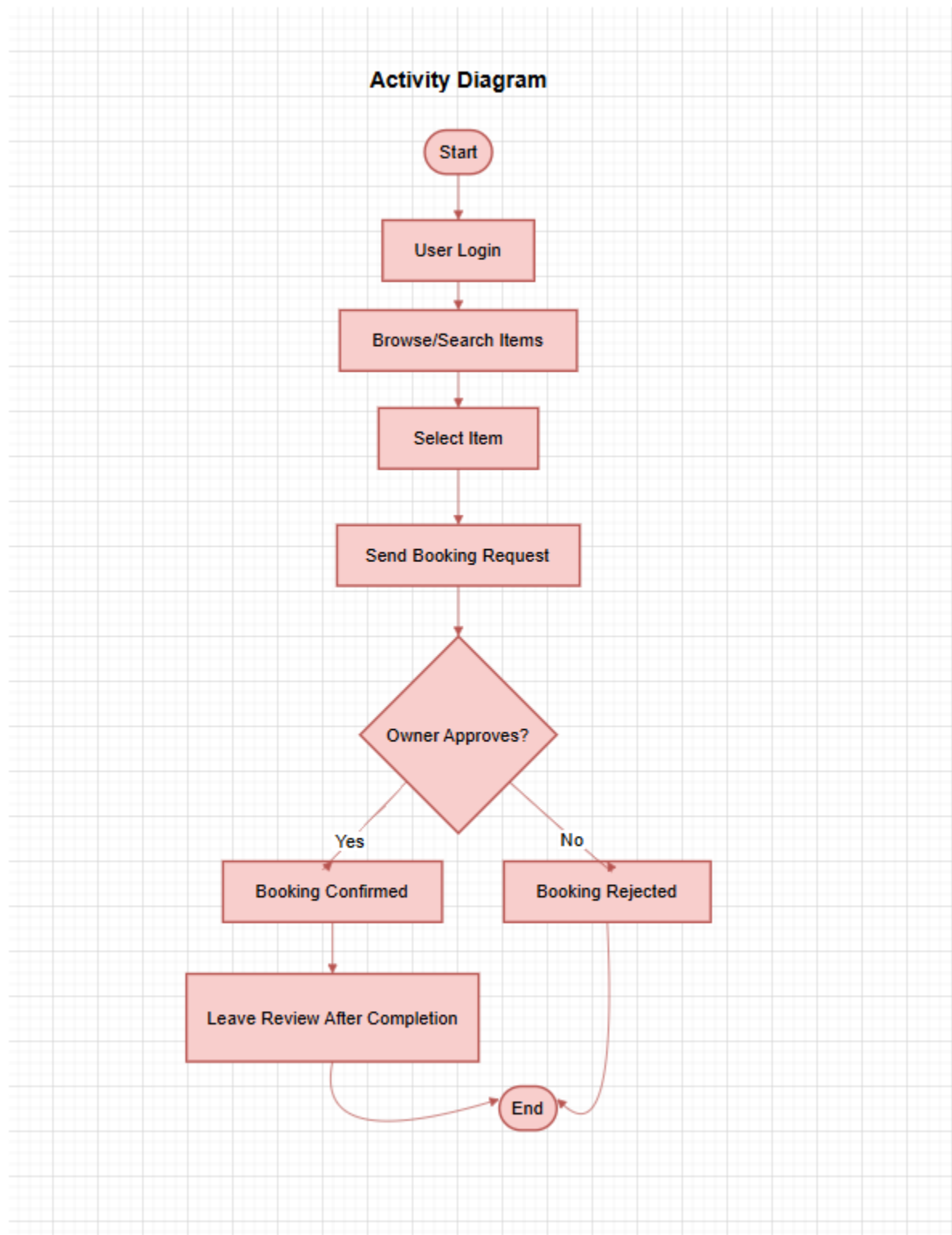
- Compliance with Bangladeshi e-commerce laws
- RESTful API standard
- Unit testing, integration testing, and UAT required

8. System Diagrams

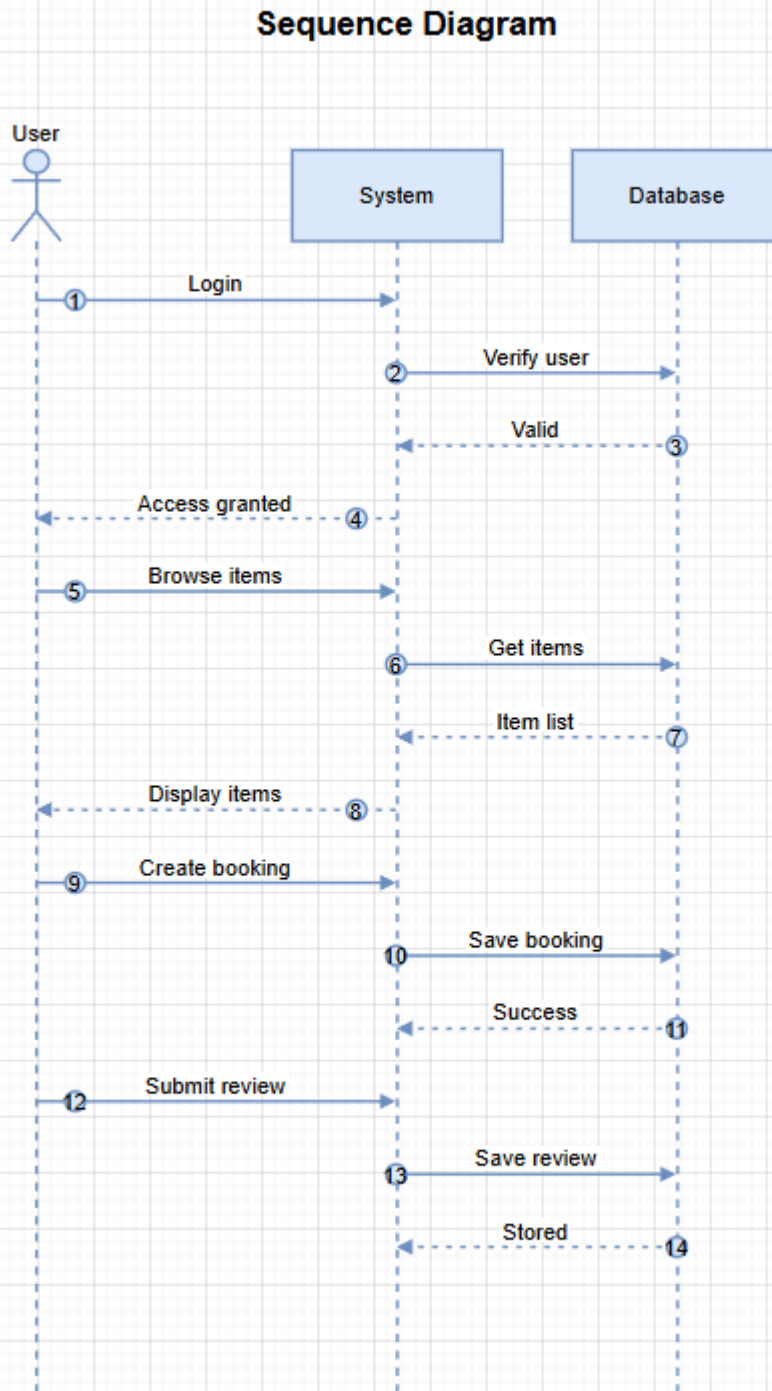
8.1 Use Case Diagram



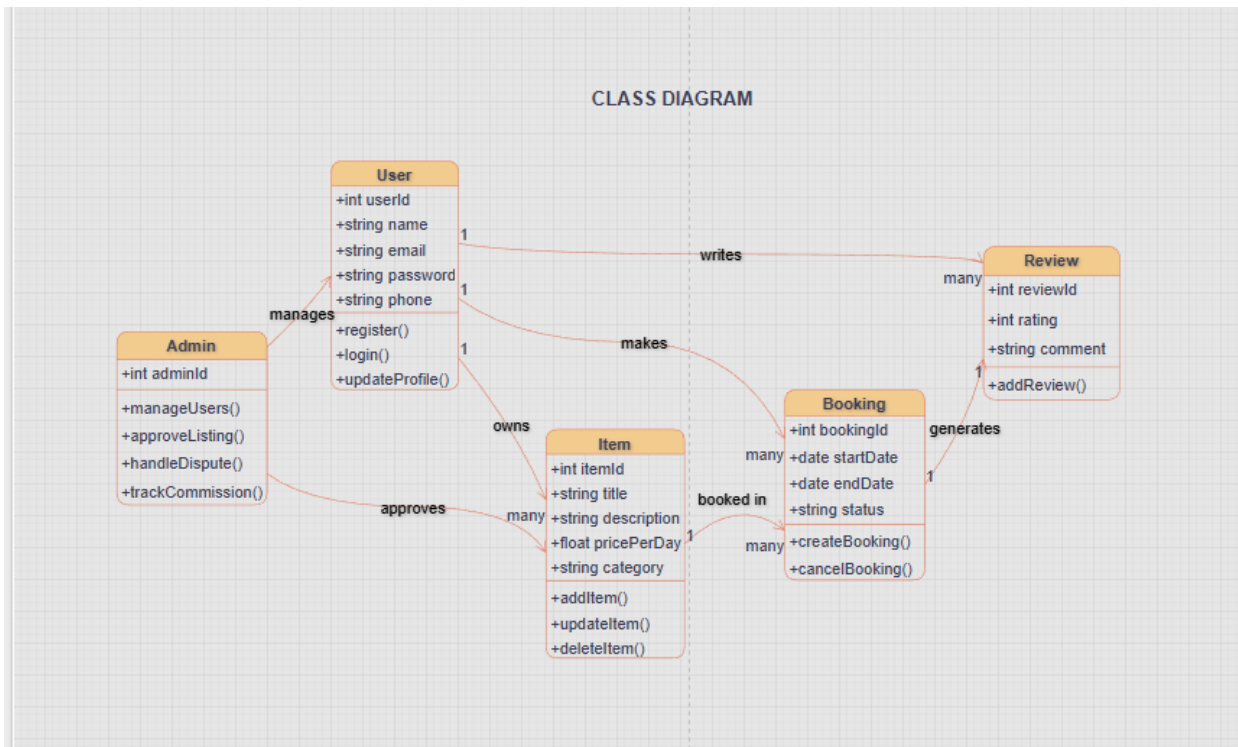
8.2 Activity Diagram



8.3 Sequence Diagram



8.4 Class Diagram



9. Conclusion

BorrowBondhu is designed as a secure, scalable, and user friendly peer to peer rental marketplace.

This SRS document defines all functional and non-functional requirements necessary to develop the MVP successfully while allowing future expansion.